Definite integrals

1 Evaluate the following definite integrals:

a $\int_0^3 (4x+5)\,dx$

b $\int_{-2}^0 (10x+4)\,dx$

c $\int_2^4 (6x+5)\,dx$

d $\int_3^4 (-4x+3)\,dx$

e $\int_{-4}^5 (-8x+3)\,dx$

f $\int_{-10}^{-6} (-5x+8)\,dx$

g $\int_{-2}^0 9x^2\,dx$

h $\int_{-1}^3 9x^2\,dx$

i $\int_{-2}^1 \left(x^2+4\right)\,dx$

j $\int_{-1}^2 \left(9x^2+1\right)\,dx$

k $\int_4^6 \left(9x^2+2x+7\right)\,dx$

l $\int_{-4}^2 x\,(x-4)\,dx$

m $\int_0^4 5x^{\frac{3}{2}}\,dx$

n $\int_{-3}^6 x\,(x+3)\,(x-6)\,dx$

o $\int_6^{11} \sqrt{x-2}\,dx$

p $\int_3^6 \left(\sqrt{x-2}+5\right)\,dx$

q $\int_3^4 (2x+3)^3\,dx$

r $\int_{-4}^2 \left((x+2)^3+3\right)\,dx$

s $\int_1^2 \dfrac{x^5-x^{-2}}{x^2}\,dx$

Properties of definite integrals

2 Consider the function $f(x)=x^2$. Find the value of the following:

a $\int_1^1 f(x)\,dx.$

b $\int_5^5 f(x)\,dx.$

c $\int_{-2}^{-2} f(x)\,dx.$

d $\int_a^a f(x)\,dx.$

3 Consider the function $f(x)=2x$.

a Find the value of $\int_0^3 f(x)\,dx.$

b Find the value of $\int_0^3 5f(x)\,dx.$

c Find the value of $5\int_0^3 f(x)\,dx.$

d State the property of definite integrals demonstrated by parts (b) and (c).

4 Given that $\int_{-2}^7 f(x)\,dx=2$, find $\int_{-2}^7 5f(x)\,dx.$

5 Given that $\int_{-1}^6 f(x)\,dx=3$, find $\int_{-1}^6 (9f(x)-2)\,dx.$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

6 Consider the function $f(x) = 6x$.

- **a** Find the value of $\int_4^8 f(x)\,dx$.
- **b** Find the value of $\int_8^4 f(x)\,dx$.
- **c** State the property of definite integrals demonstrated by parts (a) and (b).

7 Given that $\int_4^6 f(x)\,dx = 3$, find the values of the following:

- **a** $\int_6^4 f(x)\,dx$
- **b** $\int_4^6 3f(x)\,dx$
- **c** $\int_4^6 (f(x) + x)\,dx$

8 Given that $\int_{-1}^2 f(x)\,dx = 4$ and $\int_2^8 f(x)\,dx = 8$, find the values of the folowing:

- **a** $\int_{-1}^8 f(x)\,dx$
- **b** $\int_8^{-1} f(x)\,dx$
- **c** $\int_{-1}^2 2f(x)\,dx + \int_2^8 3f(x)\,dx$

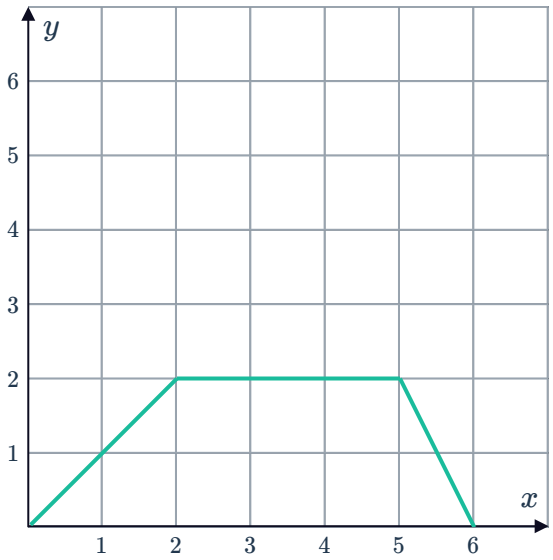
9 Given that $\int_{-1}^3 f(x)\,dx = 5$ and $\int_2^3 f(x)\,dx = 2$, find the values of the following:

- **a** $\int_{-1}^2 f(x)\,dx$
- **b** $\int_3^{-1} f(x)\,dx$
- **c** $2\int_{-1}^2 f(x)\,dx + \int_2^3 3f(x)\,dx$

Signed areas

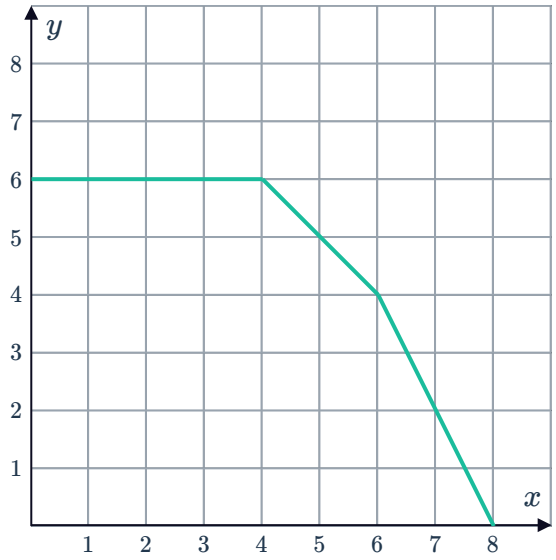
10 Consider the graph of $y = f(x)$.

- **a** Find the value of $\int_0^2 f(x)\,dx$.
- **b** Find the value of $\int_2^5 f(x)\,dx$.
- **c** Find the value of $\int_5^6 f(x)\,dx$.
- **d** Hence state the area bounded by the function and the x-axis.
- **e** Write a single definite integral to represent the area bounded by the function and the x-axis.



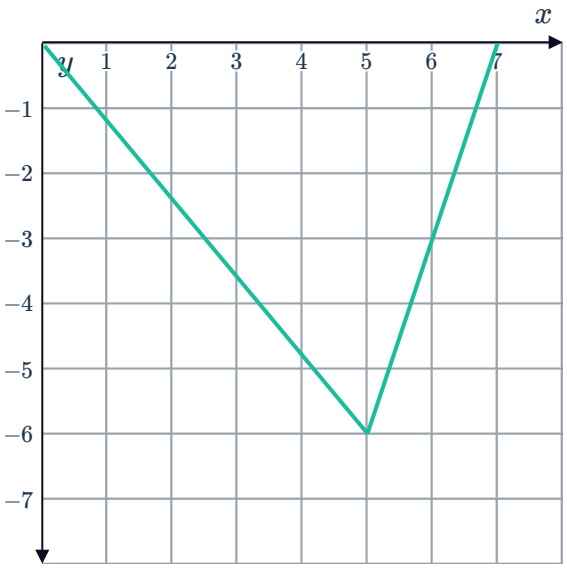
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 Consider the graph of $y = f(x)$.



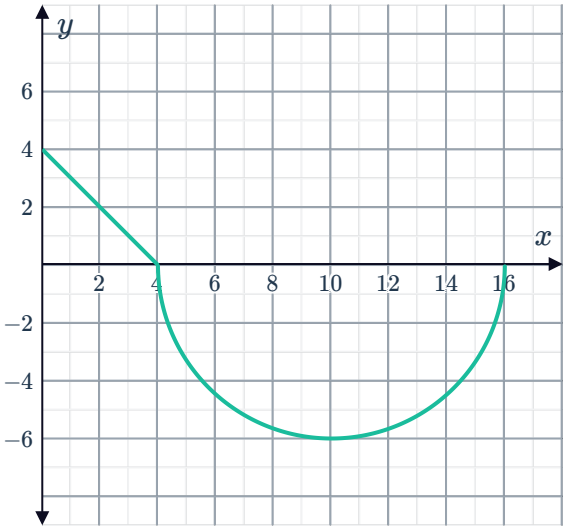
- **a** Find the value of $\int_0^4 f(x)\,dx$.
- **b** Find the value of $\int_4^6 f(x)\,dx$.
- **c** Find the value of $\int_6^8 f(x)\,dx$.
- **d** Hence state the area bounded by the function and the x-axis.
- **e** Write a single definite integral to represent the area bounded by the function and the x-axis.

12 Consider the graph of $y = f(x)$.



- **a** Find the value of $\int_0^5 f(x)\,dx$.
- **b** Find the value of $\int_5^7 f(x)\,dx$.
- **c** Hence find the value of $\int_0^7 f(x)\,dx$.
- **d** State the area bounded by the function and the x-axis.
- **e** Write a single definite integral to represent the area bounded by the function and the x-axis.

13 Consider the graph of $y = f(x)$.



- **a** Find the value of $\int_0^4 f(x)\,dx$.
- **b** Find the exact value of $\int_4^{16} f(x)\,dx$.
- **c** Hence find the exact value of $\int_0^{16} f(x)\,dx$.
- **d** Calculate the exact area bounded by the curve, the x-axis and the y-axis.
- **e** Write an expression for the exact area from part (d) as a sum or difference of definite integrals.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14 Consider the function $y = -5$.

a State whether the graph is above or below the x-axis.

b Calculate $\int_{-4}^{3} (-5)\ dx$.

c Hence find the area bounded by the curve, the x-axis and the bounds $x = -4$ and $x = 3$.

15 Consider the function $y = 2x - 8$.

a Find the x-intercept of the function.

b State the values of x for which the graph is above the x-axis.

c Calculate $\int_{-2}^{4} (2x - 8)\, dx$.

d Hence find the area bounded by the line, the x-axis and the bounds $x = -2$ and $x = 4$.

16 Consider the function $y = (x - 3)(x - 9)$.

a Find the x-intercepts of the function.

b State the values of x for which the graph is below the x-axis.

c Calculate $\int_{3}^{9} (x - 3)(x - 9)\, dx$.

d Hence find the area bounded by the curve, the x-axis and the bounds $x = 3$ and $x = 9$.

17 Consider the function $y = -(x + 2)(x + 8)$.

a Find the x-intercepts of the function.

b State the values of x for which the graph is above the x-axis.

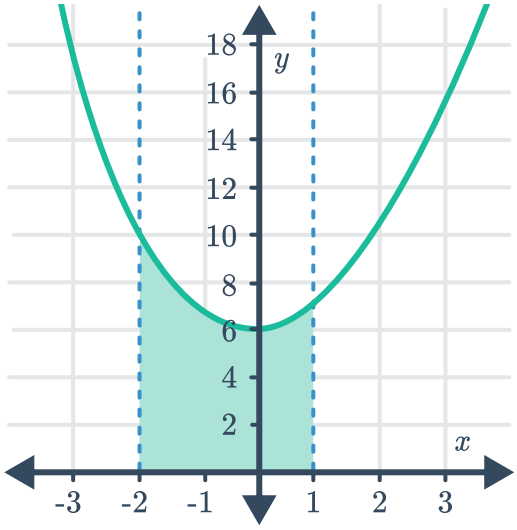
c Calculate $\int_{-8}^{-2} -(x + 2)(x + 8)\, dx$.

d Hence find the area bounded by the curve, the x-axis and the bounds $x = -8$ and $x = -2$.

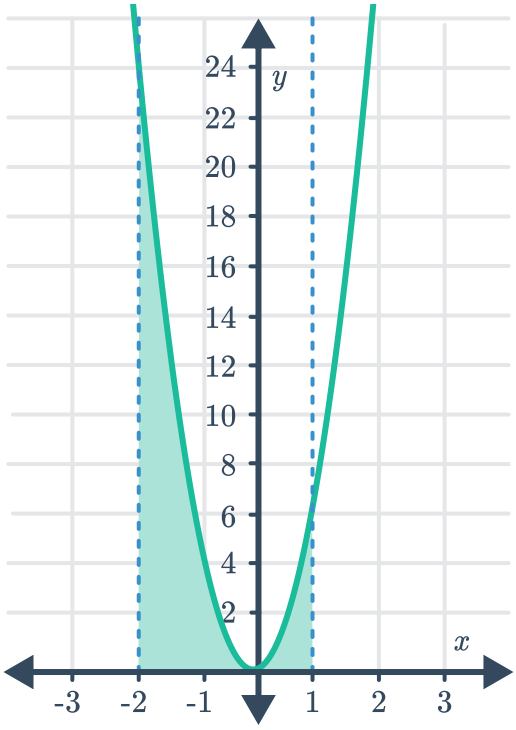
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Area under curve

18 Consider the graph of the curve $y = x^2 + 6$.
Find the exact area of the shaded region.

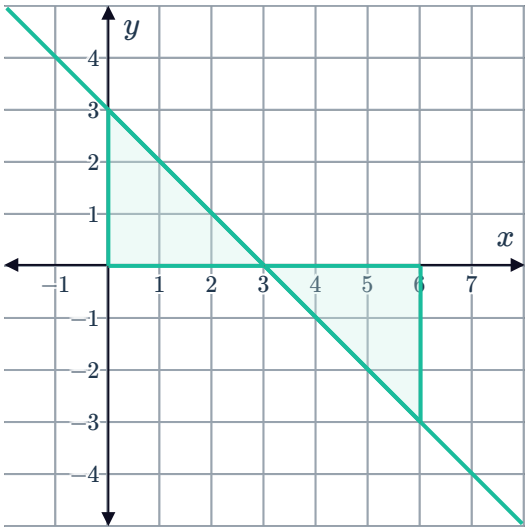


19 Consider the graph of the curve $y = 6x^2$.
Find the exact area of the shaded region.

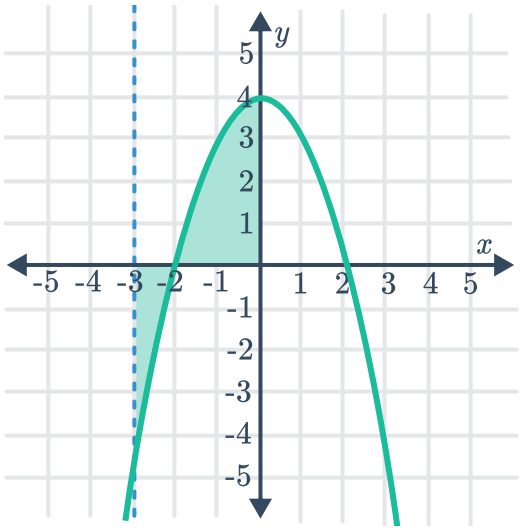


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

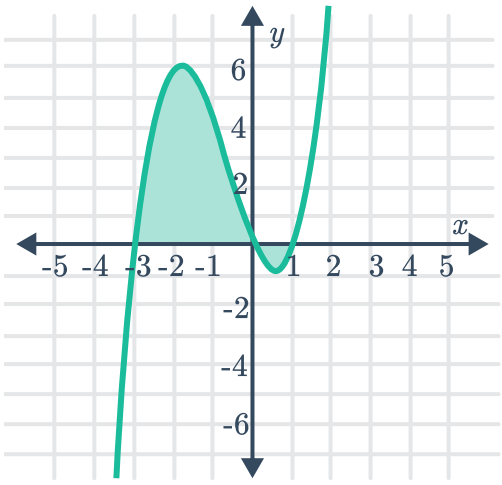
20 Consider the graph of the line $x + y = 3$.
Find the exact area of the shaded region.



21 Consider the graph of the curve $y = 4 - x^2$.
Find the exact area of the shaded region.



22 Consider the graph of the curve
$y = x\,(x - 1)\,(x + 3)$.
Find the exact area of the shaded region.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23 Consider the given functions.

i Sketch a graph of the function.

ii Hence calculate the exact area bounded by the curve, $x = 1$, $x = 3$ and the x-axis.

a $y = 2x + 3$ **b** $y = -2x + 8$ **c** $y = 4x - x^2 - 3$

24 Consider the function $y = (x - 1)^2 (x + 3)$.

a Sketch a graph of the function.

b Hence calculate the exact area bounded by the curve, x-axis and the lines $x = -1$ and $x = 2$.

25 Consider the function $y = (x + 1)^3 + 2(x + 1)^3 + 2$.

a Sketch a graph of the function.

b Hence calculate the exact area bounded by the curve, x-axis, and the lines $x = -2$ and $x = 1$.

26 Consider the function $y = \sqrt{x + 1}$.

a Sketch a graph of the function.

b Hence calculate the exact area bounded by the curve, the x-axis, and the line $x = 3$.

27 For each of the following functions:

i Sketch a graph of the function.

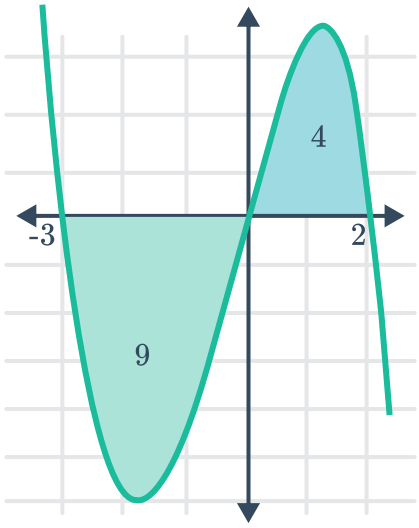
ii Hence determine the exact area bounded by the curve and the x-axis.

a $y = x(x - 1)(x + 3)$ **b** $y = (x - 1)(x + 2)(x + 3)$

c $y = x(x - 1)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

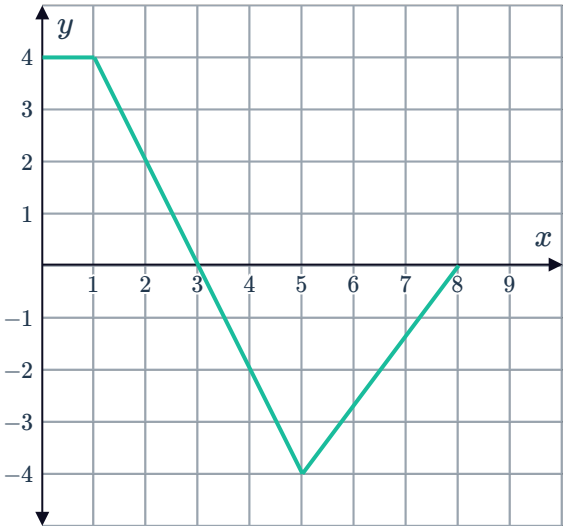
28 Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Find the value of the following:



- **a** $\int_{-3}^{0} f(x)\,dx$
- **b** $\int_{-3}^{2} f(x)\,dx$
- **c** The area enclosed by the curve and the x-axis for $x < 0$.
- **d** The area enclosed by the curve and the x-axis.

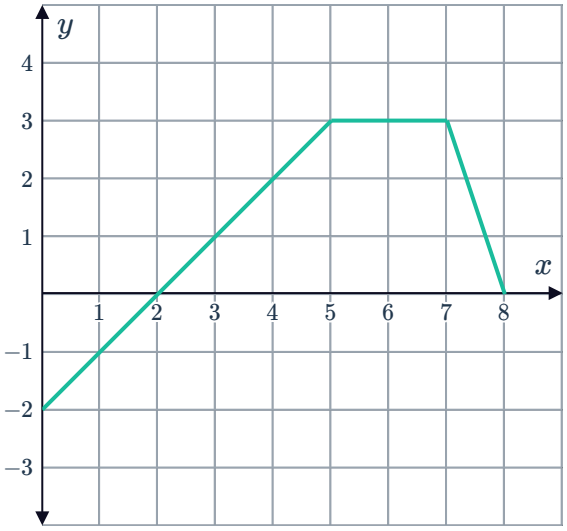
Area as an absolute value

29 Consider the graph of $y = f(x)$.



- **a** Find the value of $\int_{0}^{3} f(x)\,dx$.
- **b** Find the value of $\int_{3}^{8} f(x)\,dx$.
- **c** Hence calculate $\int_{0}^{8} f(x)\,dx$.
- **d** Calculate the area bounded by the function, the x-axis and the y-axis.
- **e** Write an expression for the area from part (d) as a sum or difference of definite integrals.
- **f** Write an expression for the area from part (d) using an absolute value sign.

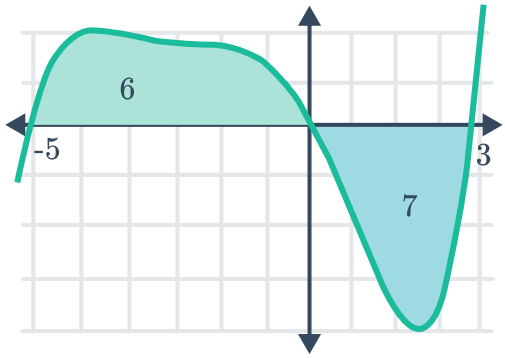
30 Consider the graph of $y = f(x)$.



- **a** Find the value of $\int_{0}^{8} f(x)\,dx$.
- **b** Calculate the area bounded by the function, the x-axis and the y-axis.
- **c** Write an expression for the area from part (b) as a sum or difference of definite integrals.
- **d** Write an expression for the area from part (b) using an absolute value sign.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

31 Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Find the following:



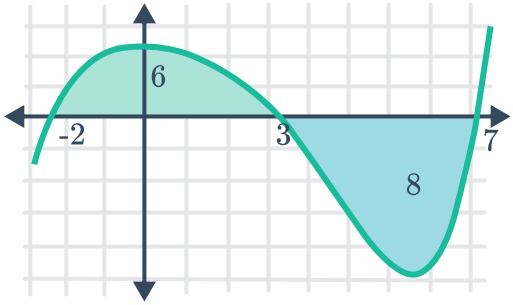
a $\int_{-5}^{3} f(x)\,dx$

b $\left|\int_{-5}^{3} f(x)\,dx\right|$

c $\int_{-5}^{3} |f(x)|\,dx$

d The area enclosed by the curve and the x-axis.

32 Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Find the following:



a $\int_{-2}^{7} f(x)\,dx$

b $\int_{3}^{7} (-f(x))\,dx$

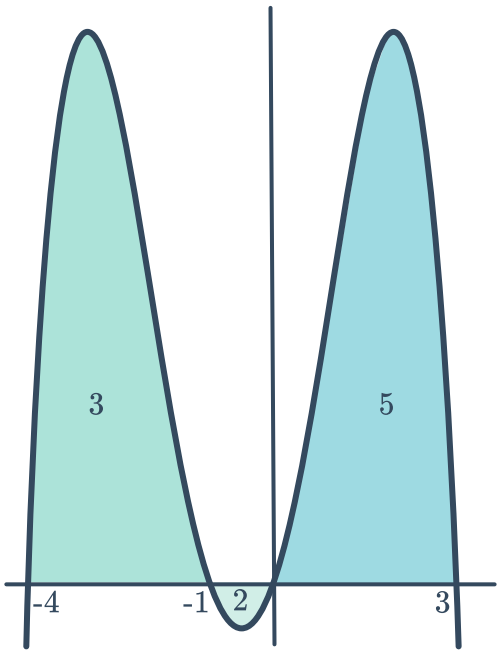
c $\int_{-2}^{7} 2f(x)\,dx$

d $\int_{3}^{-2} f(x)\,dx + \int_{3}^{7} f(x)\,dx$

e $\int_{-2}^{7} |f(x)|\,dx$

f The area enclosed by the curve and the x-axis.

33 Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Find the following:



a $\int_{-4}^{0} f(x)\,dx$

b $\left|\int_{-4}^{3} f(x)\,dx\right|$

c $\int_{-1}^{0} 2f(x)\,dx + \int_{3}^{0} f(x)\,dx$

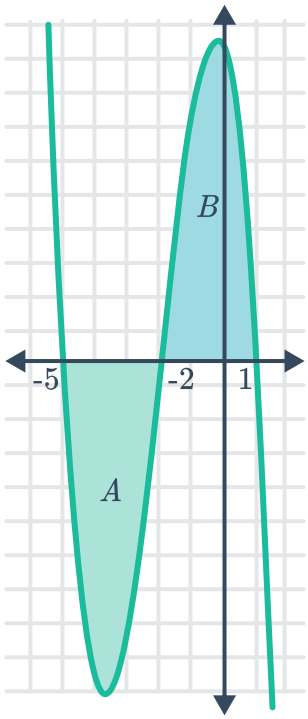
d The area enclosed by the curve and the x-axis.

e $\int_{-4}^{3} |f(x)|\,dx$

f $\int_{-4}^{3} \left(f(x) + x^2\right)dx$ given that the definite integral $\int_{-4}^{3} x^2 dx = \frac{91}{3}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

34 Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Write the following in terms of A and B:



a $\int_{-5}^{1} f(x)\,dx$

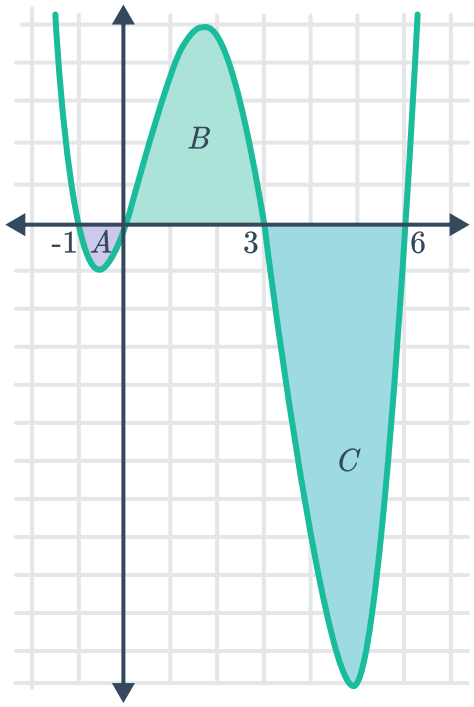
b $\int_{-5}^{-2} 3f(x)\,dx - \int_{-2}^{1} f(x)\,dx$

c $\left|\int_{-5}^{1} f(x)\,dx\right|$

d $\int_{-5}^{1} |f(x)|\,dx$

e $\int_{-5}^{1} (f(x)+x)\,dx$ given that the definite integral $\int_{-5}^{1} x\,dx = -12$.

35 Consider the function $f(x)$ as shown. The numbers in the shaded regions indicate the area of the region.
Write the following in terms of A, B and C:



a $\int_{0}^{6} f(x)\,dx$

b $\int_{6}^{-1} f(x)\,dx$

c The area bounded by the curve and the x-axis.

d $\int_{-1}^{6} |f(x)|\,dx$

e $\int_{-1}^{6} (2x - f(x))\,dx$ given that the definite integral $\int_{-1}^{6} 2x\,dx = 35$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

36 Consider the function $f(x)$ where $x = -4$, 1 and 3 are the only x-intercepts and $\int_{-4}^{1} f(x)\,dx = 4$ and $\int_{1}^{3} f(x)\,dx = -7$.
Find the following:

a $\int_{-4}^{3} f(x)\,dx$

b $\int_{-4}^{3} |f(x)|\,dx$ of $f(x)$.

c The area bounded by the curve of $f(x)$ and the x-axis.

d $\int_{3}^{-4} \left(f(x) - x^3\right) dx$, given that $\int_{3}^{-4} x^3 dx = \dfrac{175}{4}$.

37 Consider the function $f(x)$ where $x = -2$, 2 and 8 are the only x-intercepts and $\int_{-2}^{2} f(x)\,dx = -5$ and $\int_{-2}^{8} f(x)\,dx = 3$.
Find the following:

a $\int_{2}^{8} f(x)\,dx$

b The area bounded by the curve and the x-axis.

c $\int_{-2}^{8} (2f(x) - 6x)\ dx$, given that $\int_{-2}^{8} x\,dx = 30$.

38 Consider the function $f(x)$ where $x = -6$, -2, 2 and 7 are the only x-intercepts and $\int_{-6}^{-2} f(x)\,dx = -A$, $\int_{-2}^{2} f(x)\,dx = B$ and $\int_{2}^{7} f(x)\,dx = -C$.
Write the following in terms of A, B and C:

a $\left|\int_{-6}^{7} f(x)\,dx\right|$

b The area bounded by the curve and the x-axis.

c $\int_{-2}^{7} f(x)\,dx - \int_{-6}^{2} f(x)\,dx$

d $\int_{-6}^{2} 2f(x)\,dx + \int_{2}^{7} \dfrac{f(x)}{2} dx$

e $\int_{-6}^{7} (3 - f(x) + x)\,dx$, given that $\int_{-6}^{7} (3 + x)\,dx = \dfrac{91}{2}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647